

Matthew Frieri

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EDUCATION

University of Toronto | *CGPA: 3.95/4*

Graduating Spring 2028

- Computer Science Major and Data Science Specialist
- Focus in Technology Leadership. Chosen as one of 30 from 500+ applicants.

EXPERIENCE

ML Engineer | *RBC Capital Markets*

May – Dec 2025

- Developed a **neural network** to predict volume in the global equities market. Consolidated multiple legacy models, expanding inference coverage by **43% more stocks worldwide**, while **reducing aggregate error by 21%**.
- Querried time series data in kdb+/Q and conducted **feature engineering** in Python by analyzing feature/target distributions, applying appropriate normalizations, and creating meaningful categorical embeddings.
- Built a Streamlit dashboard to monitor feature importance in real-time, supporting model refinement through the ablation of 159 features, **reducing model size by 64%** while maintaining symmetric and asymmetric error.

Software Developer | *RBC Capital Markets*

Jan – August 2026

- Developed a pub-sub framework to serve real-time inference of the volume prediction model mentioned above. The signal is used by RBC's algorithmic trading platform, *Aiden*, which trades **~\$1.54B of daily notional value**.
- Configured **Prometheus** and **Loki** data sources to create a **Grafana** dashboard for the signal, monitoring live performance and alerting unwanted behavior via email. Critical to confidently bring the new model to production.
- Developed a data source "parallelizer" to abstractly read/write/process data via multi-threading and multi-processing from various sources to sinks (parquet, arrow, s3, kdb+). Queried data up to **53% faster in parallel**.

Full Stack Developer | *MultiCloud Decentralized Applications Inc.*

July – Dec 2024

- Developed and designed the front end of the company's website using **Next.js** and **Tailwind CSS**, employing modular design and reusable components to streamline development and significantly boost efficiency.
- Implemented backend features including a tree based file system, file search, and an automated email service, helping bring the platform closer to production readiness.
- Enhanced **SEO** and internal website performance using Google Lighthouse, realizing a **40% increase** in web crawler visibility, loading times, and accessibility.

Undergraduate Research Lead and Mentor | *Youreka Canada w/ UofT*

Jan – May 2024

- **Led a team of 4** junior researchers through the analysis of large datasets to reveal statistically significant trends.
- Applied linear regression techniques using **Python** to uncover trends in ADHD rates among children.
- Mentored junior researchers, fostering collaboration and team bonding. Created a supportive and innovative environment that boosted research quality and promoted personal growth among the junior researchers.

PROJECTS

Cash App – Exposing Bias in Fraud Detection 🚫 | *Python, Statsmodels, Flask, React* Sept – Dec 2024

- Worked with a team of 5 in collaboration with Cash App to develop a dashboard that reveals bias in their AI fraud detection system, identifying disproportionate targeting through detailed, interpretable visualizations.
- Trained an **ARIMA** model to predict the impact of removing specific biases from Cash App's fraud detection system, estimating a **20% reduction in false positives** for underrepresented user groups.

Impactify (1st Place @ Hack The Valley) 🚫 | *Esp32, Django, PostgreSQL, React, Twilio*

Oct 2024

- Embedded hardware in a hockey helmet to detect high impact collisions so coaches can monitor their players live.
- Developed a live dashboard for coaches and players, where data is streamed from the helmet.
- Implemented automatically sending SMS notifications to coaches when a player experiences a traumatic impact.

TECHNICAL SKILLS & INTERESTS

Programming Languages: Python, C, kdb+/Q, Java, C#, SQL, JavaScript, TypeScript, HTML/CSS

Technologies: Git, PyTorch, MLFlow, Pandas, Polars, Numpy, S3, React, Django, FastAPI, OpenCV, MongoDB

Interests: Outdoor Bouldering, Backcountry Canoe Camping, Tennis, Table Tennis, Self Hosting / Home Server